Oluwaseun A. Adekoya

Robotics Simulation & Physical AI Engineer | 6+ Years of Engineering Experience
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PROFESSIONAL PROFILE

Robotics engineer developing GPU-accelerated digital twins, AI Agents and learned robot policies, with hands-on benchmarking of motion planners, perception models, and frontier/open AI systems. Experience with NVIDIA Isaac Sim/Isaac Lab, MuJoCo, Python, PyTorch/CUDA, Gazebo, ROS 2 and physical robot experimental validation.

FEATURED ROBOTICS SIMULATION & PHYSICAL AI PROJECTS

AI-Agent Autopipetting & Dual-Arm Co-Manipulation | Seeed reBot Arm B601-RS
MuJoCo digital twin | AI Agents / LLM-based task planning | RL policies | simulated cameras / hardware RGB-D

- Built a physics-based laboratory digital twin and tested robotic tube pick-and-place, complete pipetting, teleoperation, and dual-arm grasping, twisting, cap opening, and insertion; also deployed manipulation on physical reBot arms.
- Developed LLM-powered AI Agents that translate liquid-volume requests and tagged well IDs into executable pipetting sequences using simulated camera and barcode cues.
- Benchmarking frontier AI models via OpenAI and Gemini APIs for task-plan validity, robotic task success, response latency, hallucination/error rates and token cost (ongoing).
- Trained RL policies for dual-arm grasping, twisting and cap opening; created multi-view technical demonstrations of robotic task execution.

CNC Machine-Tending Digital Twin & Collision Avoidance | Yaskawa Motoman HC10 (6-DOF)
NVIDIA Isaac Sim + Isaac Lab | PRM / speed-and-separation monitoring (SSM) | physical validation

- Developed an Isaac Sim/Isaac Lab digital twin of a CNC-to-conveyor tending cell and evaluated safety-aware motion planning in repeatable workcell scenarios; validated planning on the physical Yaskawa Motoman HC10.
- Achieved >98.5% speed-and-separation compliance; accelerated collision checking to 10–11 ms per 50 samples and demonstrated ~6 s look-ahead safety planning in Ohio BWC-funded research.
- Benchmarked HAMP, MARSHA, TPRM and RRTConnect for goal success, SSM compliance, planning/solve time and robot execution time.

Vision-Guided Robotic Fastener Removal | Neuromeka Indy RP2 v2 (7-DOF)
MuJoCo digital environment | 3D point clouds | eye-in-hand RGB-D | 6-DoF pose estimation

- Built the MuJoCo robotic screwdriving test digital environment to evaluate alignment, bit insertion and removal for Phillips, hex-head and hex-socket fasteners on challenging surfaces.
- Benchmarked AprilTag, SAM6D and FoundationPose for visual localization and pose estimation; contributed to a team study achieving 91% success in >800 physical fastener tests.
- In related RL-based screwdriving force-control experiments, integrated reinforcement learning with impedance control (98.64% task success).

RL Waypoint Planning with Nonlinear MPC Tracking | Neuromeka Indy RP2 v2
MuJoCo physics simulation | reinforcement learning | NMPC / acados | collision avoidance

- Trained GPU-accelerated TD3 RL waypoint policies using PyTorch/CUDA (89.5% goal-reaching success on 200 held-out simulated scenarios); used NMPC/acados to track collision-avoiding waypoints in MuJoCo.

TECHNICAL SKILLS

Simulation & digital twins: NVIDIA Isaac Sim, Isaac Lab, MuJoCo, Gazebo, MATLAB/Simulink, Simscape, URDF/CAD integration
Physical AI, robotics & vision: AI Agents, LLM task planning, frontier/open-model benchmarking, reinforcement learning, dual-arm manipulation, kinematics/coordinate transforms, RGB-D, point-cloud vision, PRM/OMPL, MoveIt, SSM, NMPC, impedance control
Programming & tools: Python, C++, PyTorch, CUDA, TensorFlow, JAX, ROS 2, Linux/Ubuntu, Git/GitHub, Docker, CasADi, acados; Autodesk Fusion 360, SolidWorks, Siemens NX

SELECTED FUNDED RESEARCH

- Ohio BWC (2026): safety-aware robotics; UC Office of Research (2024): MPC vs. DP benchmarking; Industry 4.0/5.0 / Digital Futures (2023): PHEV control co-design.

PROFESSIONAL EXPERIENCE

Center for Intelligent Metrology and Sensing, University of Cincinnati | Industry Research Engineer *Mar 2026 – Present*

- Developed a CNC machine-tending digital twin in NVIDIA Isaac Sim/Isaac Lab and safety-aware planning algorithms for the Yaskawa HC10; achieved >98.5% speed-and-separation compliance in Ohio BWC-funded research.
- Optimized Python collision checking to 10–11 ms per 50 samples, implemented ~6 s look-ahead safety planning, and validated robot motions through simulation and physical HC10 deployment.
- Benchmarked RGB-D safety-monitoring models including ConvNeXtV2, ResNet-50, HR48, ViT, RegNet-Y, Gemini and open Gemma 3 12B; built MuJoCo environments for Indy RP2 robotic fastener removal.

Procter & Gamble, Mason, OH | Production Systems & Reliability Engineering Intern *Aug 2024 – Dec 2025*

- Modeled reliability of six DayQuil/NyQuil production lines; identified dominant failure modes and increased sustained process reliability from 87.1% to 92.3%.
- Developed a comparable production-system model to support startup planning for a new manufacturing plant in Greensboro, NC.

Integrated Vehicle Design Laboratory, University of Cincinnati | Graduate Researcher *Jan 2021 – Aug 2024*

- Led design and evaluation of real-time nonlinear MPC for plug-in hybrid electric vehicle powertrains in Industry 4.0/5.0 research supported by industrial partners.
- Benchmarked MPC against dynamic programming: $0.027/mile MPC fuel cost vs. $0.004/mile DP and $0.032/mile prior work; reduced solve time from ~6 min to 4.28 min.

Mikano International Ltd., Lagos, Nigeria | Associate Lead Engineer *Apr 2019 – Dec 2020*

- Supervised four generator assembly rows in an ~80-unit/3-day production operation; led assembly, maintenance, quality checks and SOP reviews for 9–2,000 kVA systems.

British American Tobacco, Ibadan, Nigeria | Assistant Project Engineer *Aug 2017 – Jan 2018*

- Supervised piping installation for two industrial desiccant dryers and produced the 2D CAD drawings used for commissioning.

EDUCATION

Ph.D., Mechanical Engineering | University of Cincinnati, OH *Expected May 2027*
Dissertation: Scene Graph-Guided LLM Task Planning and Reinforcement Learning for Safe Robotic Manipulation
M.S., Mechanical Engineering | University of Cincinnati, OH *Dec 2024*
B.Eng., Mechanical Engineering | FUTA, Nigeria (First Class; Best Graduating Student) *Nov 2018*

SELECTED PUBLICATIONS & MANUSCRIPTS

- **Adekoya, O.**, Pan, Q., Onaga, K., Yahia, T., Li, R., Dong, J., Jia, X. Critical Distance-Aware Collision Avoidance Using Probabilistic Roadmaps (PRM) and Human-Aware Motion Planners (HAMP). (Under Review).
- Yahia, T., Pan, Q., Onaga, K., Haubeil, K., **Adekoya, O.**, Ma, O., Dong, J., Jia, X. Experimental Evaluation of RGB-D Sensing for Speed and Separation Monitoring in Human-Robot Collaboration Work Cells. *Robotics and Computer-Integrated Manufacturing* (Under Review).
- Suer, A. D., Pan, Q., Yahia, T., Marasini, G., Onaga, K., **Adekoya, O.**, Haubeil, K., Dong, J., Jia, X. *Novel Methodology for 3D Vision-Guided Robotic Fastener Removal.* Measurement (Under Review).
- **Adekoya, O.**, Alexander-Ramos, M. *A Comparative Study Between Dynamic Programming and Model Predictive Control for Closed-Loop Control Systems.* Journal of Mechanical Design, 2024.

AWARDS, FELLOWSHIPS & HONORS

- Three-time NSBE BCA/Affiliate/Fellows Scholarship recipient (2023, 2024, 2026); NSBE R4 Tinkered Out! Scholarship (2024).
- Paige-Lynn Roberts Memorial (Riedman) Scholarship (2025); Graduate Student Engineer of the Month, University of Cincinnati (Dec 2024).
- University Research Council Fellowship, PEWS Engaging Science Fellowship, and NSF Future of Semiconductors Fellowship (all 2024).
- Best Graduating Student, Mechanical Engineering, FUTA; Vice Chancellor's First-Class Award (2018).

CERTIFICATIONS & TRAINING

Lean Six Sigma Yellow Belt | Python, MATLAB and Simulink training | Machine Learning Onramp | Technical Writing